

Neonatal Sepsis

Contents

Neonatal Sepsis	2
Outline	2
Introduction	2
Burden	2
The definition problem	3
Definitions	3
Diagnostic categories	3
SIRS – how relevant to neonatal sepsis?	3
SIRS criteria – presence of 2 of the following	4
Severe sepsis and septic shock	4
Epidemiology	4
Types of neonatal sepsis	5
In developed countries (3 types)	5
Early versus late onset	5
In developing countries	5
Aetiology	5
Developed countries	5
Developing countries	5
Risk and host factors	5
Maternal – MAJOR	5
Fetal / neonatal	6
Environmental	6
MINOR risk factors	6
Labour characteristics	6
Infant characteristics	6
Stages of sepsis development – the seed-soil principle	7
Susceptibility of the newborn to infection	7
Diagnosis	7
Clinical features	7
Community diagnosis of sepsis	8
Suspicious clinical signs	8
Criteria for diagnosis – clinical variables	8
Haemodynamic and tissue perfusion variables	9
Differential diagnosis	9
Principles of management	9
Investigations	9
Sepsis screen	9

Complete blood count – findings suggestive of sepsis	10
Urine	10
Töller score	10
Other ancillary investigations	10
Other laboratory findings	10
Other investigations	10
Blood culture and other samples	11
Superficial swab cultures	11
Antibiotic therapy	11
Supportive management	11
Other ancillary treatments	12
Prevention	12

Neonatal Sepsis

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Outline

Introduction and definitions · Epidemiology · Aetiology · Risk factors and host factors · Clinical features · Differential diagnosis · Investigations · Treatment · Complications · Prognosis · Prevention

Introduction

Neonatal sepsis is **one of the leading causes of neonatal mortality** and a **major public health problem**, especially in developing countries.

Even though recent medical advances have improved neonatal care, many challenges remain in the diagnosis and management of neonatal infections.

The diagnosis is complicated by:

- The frequent presence of **non-infectious conditions that resemble sepsis**, especially in preterm infants (e.g. RDS, CHD)
- The **absence of optimal/definitive diagnostic tests**
- **Viral and fungal infections** that occur in newborns and **mimic bacterial infections**

Since neonatal sepsis is a high-risk disease, especially in preterm infants, clinicians are compelled to **empirically administer antibiotics** to infants with risk factors and/or signs of suspected sepsis – the **“RULE OUT SEPSIS SYNDROME” (ROSS)**.

Burden

- NNS is the **3rd leading cause of neonatal mortality**, after **asphyxia** and **LBW/prematurity**
- The **top 4 diagnoses** in most newborn units are **asphyxia, NNS, preterm/LBW and jaundice**
- **13%** of all neonatal mortality
- **42%** of deaths in the first week of life
- Many cases are **not reported**, especially in developing countries
- NNS explains about **a quarter (25%) of neonatal deaths in Nigeria**

The definition problem

- **Lack of a widely accepted and consensus definition**
- Definition relates to **clinical features and outcome of investigations**
- Lack of definitive accepted criteria that **cut across term and preterm** manifestations
- Both **broad-spectrum antibiotics** and **prolonged empirical treatment** are associated with **adverse outcomes**

Risks of overuse of antibiotics:

- Increased **antibiotic resistance**
- Negative effects of **altering the microbiome**
- Increased rate of serious **necrotising enterocolitis**
- Increased **mortality**

Definitions

Bacteraemia – the presence of **viable bacteria in the blood**.

Septicaemia – **systemic infection confirmed by the presence of actively multiplying bacteria in the blood**.

Neonatal sepsis (NNS) – a **clinical syndrome characterised by systemic signs of infection** (the infant is looking ill), **accompanied by bacteraemia, in the first 28 days of life**.

Diagnostic categories

Suspected sepsis – regardless of whether there is a clinical symptom or not, the **presence of sepsis risk factors** in the baby, or findings suggesting sepsis on follow-up.

Clinical sepsis – clinical and laboratory findings are present, but there is **failure to show the causative microorganism**.

Proven sepsis – clinical and laboratory findings are present, **plus demonstration of the pathogenic microorganism in cultures taken from a sterile field**; or a **positive blood culture or PCR** in the presence of clinical signs and symptoms of infection.

For **coagulase-negative sepsis**: **two positive blood cultures, or one positive blood culture plus a positive CRP**.

Probable sepsis – presence of signs and symptoms of infection **plus at least two abnormal laboratory results** when blood culture is negative.

Possible sepsis – presence of clinical signs and symptoms of infection **plus raised CRP or IL-6/IL-8 level** when blood culture is negative.

SIRS – how relevant to neonatal sepsis?

SIRS (Systemic Inflammatory Response Syndrome) is the **physiologic response to inflammation from any source, including infection**.

So: **Sepsis = SIRS + focus of infection**.

BUT many neonates with sepsis **lack a specific focus and signs**, especially preterms. However, SIRS still finds its use.

Strictly, diagnosis of SIRS requires that **temperature or leukocyte abnormalities** be present.

SIRS criteria – presence of 2 of the following

At least **two** of the following four criteria, **one of which must be abnormal temperature or leukocyte count**:

1. Core temperature of **> 38.5 °C or < 36 °C**
2. **Tachycardia** – mean heart rate **2 SD above normal for age** in the absence of external stimulus
3. **Bradycardia** – mean heart rate **< 10th centile** (provided no external vagal stimulus, beta-blockers, or CHD), or unexplained bradycardia of **> 30 min**
4. **Mean respiratory rate 2 SD above normal for age** (40-60 c/min)
5. **Leukocyte count elevated or depressed for age, or 10% immature neutrophils**

SIRS has not made early diagnosis of NNS realizable, because of the absence of specific symptoms and the suboptimal diagnostic value of the laboratory findings.

Severe sepsis and septic shock

Severe sepsis = sepsis plus one of:

- **Cardiovascular organ dysfunction**
- **Acute respiratory distress syndrome (ARDS)**
- **>= 2 organ dysfunctions** (respiratory, renal, neurologic, haematologic, or hepatic)

Or: sepsis + **signs of end organ damage OR hypoperfusion.**

Signs of end organ damage:

- Low PaO₂/FiO₂ (acute lung injury)
- Lack of bowel sounds (ileus)
- Oliguria
- Increased total bilirubin
- Thrombocytopenia

Signs of hypoperfusion / cardiovascular dysfunction:

- Increased lactic acid
- Low blood pressure
- Decreased capillary refill
- Change in mental status

Septic shock = severe sepsis plus sepsis-induced hypotension that is refractory to fluid resuscitation.

Epidemiology

Incidence varies:

- Across regions
- Across centres within the same region
- From time to time within the same centre

Other points:

- **Commoner in developing than developed nations**
- **Worse in preterms than terms**
- **Prevalent organisms and antibiotic sensitivity also vary from time to time**

Types of neonatal sepsis

In developed countries (3 types)

1. **Early onset NNS** – symptoms < **72 hours** (various cut-offs 48 hr - 7 days, but most epidemiological studies use 72 hours)
2. **Late onset NNS** – cases diagnosed on the **4th-30th days of life**
3. **Very late onset NNS** – cases diagnosed in infants hospitalised in the NICU from the first 30 days of life until discharge

Early versus late onset

- **Early onset** disease pathogens are related to the **maternal genital tract**, with **Group B Streptococcus (GBS)**, and the delivery area
- **Late onset** disease pathogens reflect **contamination at a later date and nosocomial infection**

In developing countries

Unlike in developed nations, in developing nations **similar organisms are usually found before and after 72 hours**.

- Most deliveries at home, by TBAs, and even in maternity homes/centres, are **infected with community-acquired pathogens**, not just maternal pathogens
- Most deliveries also occur in **unclean areas with unclean delivery practices**, leading to infections with **nosocomial organisms early in life**

Hence the classification into:

- **Community acquired**
- **Hospital acquired**

Aetiology

Developed countries

Early onset sepsis (EOS): Group B Streptococcus **43-58%**, *E. coli*, other gram-negative bacteria, *S. aureus*, coagulase-negative *Staphylococcus*, *L. monocytogenes*

Late onset sepsis (LOS): coagulase-negative *Staphylococcus*, *E. coli*, *Klebsiella* sp., *S. aureus*, *Candida albicans*, *Enterococcus* sp., *P. aeruginosa*, other *Candida* sp.

Developing countries

Community acquired: *Klebsiella* sp., *S. aureus*, *E. coli*, Group B Streptococcus, *S. pneumoniae*, *Salmonella* sp.

Hospital acquired: *Klebsiella* sp., *S. aureus*, *E. coli*, coagulase-negative *Staphylococcus*, *Pseudomonas* sp., *Enterobacter* sp., *Candida* sp.

Risk and host factors

Maternal – MAJOR

- **ROM > 18 hr / PROM**

- **Intrapartum maternal fever** ($> 38.0\text{ }^{\circ}\text{C}$ / $> 100.4\text{ }^{\circ}\text{F}$)
- **Chorioamnionitis**
- **Malodorous liquor**
- **Sustained fetal heart rate** > 160 beats/min

Fetal / neonatal

Congenital malformations, and **preterm/LBW especially VLBW and ELBW**:

- Ectopia vesicae
- Neural tube defects
- Cutis aplasia
- Exomphalos / omphalocele / gastroschisis
- Obstructive problems (GIT, GUT, CVS, etc.)

Environmental

Home, TBA, road/taxi/casualty deliveries; nursery and environmental circumstances and practices (poor soap and water use, poor spacing); **overcrowding; poor, inadequate or infrequent hand washing.**

MINOR risk factors

- Peri/intrapartum maternal fever ($> 37.5\text{ }^{\circ}\text{C}$ but $< 38.5\text{ }^{\circ}\text{C}$)
- Twin gestation
- Prematurity (< 37 weeks)
- Maternal WBC $> 15,000$
- **ROM > 12 hr**
- Tachypnoea (< 1 hour) – increased RDS is symptomatic
- **Maternal GBS colonization**
- Low APGAR (< 5 at 1 min)
- Mode of delivery
- Instrumentation and manipulations
- Family history of a sibling with systemic bacterial disease under 3 months of age
- Urinary tract infection
- Maternal substance abuse
- History of recurrent abortion
- Poor maternal nutrition

Labour characteristics

- Preterm labour
- Premature or prolonged time between rupture of membranes and delivery
- Fetal tachycardia without maternal fever, blood loss, hypotension, or tachycardia-inducing medication
- Foul lochia
- Rupture of membranes > 12 hours

Infant characteristics

- **Apgar < 6 at 5 minutes**

- Meconium-stained amniotic fluid
- Oxygen requirement
- Respiratory distress
- Neutropenia
- **Male sex**
- Congenital anomalies that cause breakdown of anatomic barriers to infection
- Polymorphonuclear leucocytes and intracellular organisms in gastric aspirate

The above portend increased risk of systemic bacterial infections in the newborn infant during the **first seven days of life**.

Stages of sepsis development – the seed-soil principle

Skin/mucosa contamination -> invasion of skin/mucosa -> multiplication of organisms -> demonstration of immunocompetence and antibody responses

Susceptibility of the newborn to infection

Both **specific and non-specific** mechanisms are **poorly developed**.

Non-specific mechanisms:

- **Skin and mucosal barriers are weak and fragile**
- **Neutrophil and monocyte response (chemotaxis) is slow and inadequate**
- **Poor opsonin support**
- **Hypocomplementaemia**
- Intracellular killing of ingested bacteria is weak

Deficient host immunity – poor intracellular killing mechanisms:

- **Oxygen-dependent mechanisms** (OH⁻, O² formation) are poorly developed
- **Non-oxygen dependent mechanisms** are also poorly developed (lysosomal enzymes)
- **Specific immunity – humoral and cellular – are weak and ineffective**
- **Relative hypo-immunoglobulinaemia**

Diagnosis

- Specific and non-specific signs and symptoms
- **Have a high index of suspicion**
- **Sepsis score**
- **SOFA score**
- **Töller sepsis score**
- Paediatric Committee (PDCO) of the European Medicines Agency (EMA) score
- **Sepsis calculators for EOS**

Clinical features

- Symptoms and signs can be **rapidly progressive**
- **Easily involves multiple organs/systems**
- Early clinical signs are **non-specific, non-localizing, subtle**, do not distinguish among organisms, and are often not organ specific – e.g. meningitis presenting with respiratory distress or diarrhoea

General: fever, temperature instability, poor colour, dusky colour, “not doing well”, poor feeding, sclerema, oedema

Gastrointestinal: poor feeding, poor swallowing/sucking, abdominal distension, vomiting, diarrhoea, water brash/excessive salivation and secretions, feed intolerance, increased feed aspirate, hepatomegaly

Respiratory: respiratory distress; increased nasal secretions; dyspnoea (flaring, chest retractions/recession); tachypnoea; bradypnoea/apnoea; noisy breathing (worsening stridor, grunting, wheezing); cheek puffing; cyanosis

Renal: oliguria, haematuria, dribbling

Cardiovascular: pallor, dusky colour, mottling; cold clammy skin; tachycardia, bradycardia, cyanosis, hypotension, oedema, dysrhythmias

Central nervous system: poor activity, weakness, excessive sleep or poor sleep, apnoea, lethargy, irritability, tremors, seizures, reduced primitive reflexes (poor sucking, swallowing, poor rooting, abnormal Moro), hyporeflexia, hypotonia, irregular respiration, full fontanelle, high-pitched cry

Haematologic: jaundice, pallor, petechiae, purpura, bleeding, ecchymosis, splenomegaly

Metabolic: unexplained hypoglycaemia, hyperglycaemia, acidosis (respiratory, metabolic), hyponatraemia

Others: organ-specific symptoms, e.g. eye discharges and swelling/oedema; ear discharges; limb swelling; umbilical redness, foul smelling, discharges

Community diagnosis of sepsis

WHO identified **seven clinical signs** as indication for neonatal sepsis, for immediate referral:

1. Difficulty feeding
2. Convulsions
3. Movement only when stimulated
4. Respiratory rate > 60 per minute
5. Severe chest indrawing
6. Axillary temperature > 37.5 °C or < 35.5 °C
7. Cyanosis and grunting

Suspicious clinical signs

- **Petechiae and purpura** in the setting of haemodynamic instability
- Fever
- Cough
- **Hypoxaemia** in the setting of leukocytosis and pulmonary infiltrates
- **Distended tympanitic abdomen** with fever and leukocytosis associated with a perforated bowel

Criteria for diagnosis – clinical variables

- Temperature instability
- **Heart rate** >= 180 beats/min or <= 100 beats/min
- **Respiratory rate** > 40 breaths/min plus grunting or desaturations
- Lethargy / altered mental status

- **Glucose intolerance** (plasma glucose > 10 mmol/L)
- **Feed intolerance**

Haemodynamic and tissue perfusion variables

Haemodynamic:

- Blood pressure **2 SD below normal for age**
- **Systolic pressure < 50 mmHg** (newborn day 1)
- **Systolic pressure < 65 mmHg** (infants ≤ 1 month)

Tissue perfusion:

- **Capillary refill > 3 seconds**
- **Plasma lactate > 3 mmol/L**

Differential diagnosis

NNS may **coexist with** other pathologies and may also **mimic** other abnormalities.

Neurologic: hypoxic ischaemic encephalopathy, neonatal seizure, intracranial haemorrhage

Respiratory: respiratory distress syndrome (especially preterm), aspiration pneumonia (amniotic fluid or gastric contents), tracheo-oesophageal fistula, transient tachypnoea of the newborn

Cardiovascular: congenital hypoplastic left heart syndrome and other structural diseases; persistent pulmonary hypertension of the newborn; acquired – myocarditis, hypovolaemic or cardiogenic shock

Gastrointestinal: necrotizing enterocolitis, spontaneous GI perforations

Metabolic: hypoglycaemia; adrenal disorders (adrenal haemorrhage and/or insufficiency, CAH); in-born errors of metabolism (organic acidurias, lactic acidosis, urea cycle disorder)

Haematologic: vitamin K clotting factor deficiency, neonatal purpura fulminans, immune mediated thrombocytopenia, immune mediated neutropenia, anaemia from other causes, haemolytic disorders (ABO, Rh isoimmunization, G6PD deficiency), other hereditary clotting disorders

Principles of management

1. **Review clinical features. Have a high index of suspicion. Balance between risks and features seen**
2. **Investigate and take necessary culture samples**
3. **Commence antibiotic treatment immediately**, especially with major risk factors
4. **Review treatment with clinical progress and laboratory results**
5. **Repeat investigations when necessary** – sudden deterioration may occur

Investigations

Sepsis screen

- Full blood count
- Blood cultures
- CSF examination
- Urine cultures
- Culture of other relevant body fluids or effluents

- **Chest radiograph if indicated**

Complete blood count – findings suggestive of sepsis

- White blood cell count **decreased** $< 5,000/\text{mm}^3$
- White blood cell count **increased** $> 34,000/\text{mm}^3$
- **Absolute neutrophil** $< 1,000/\text{mm}^3$
- **Bands to total neutrophil count ratio** > 0.2
- **Immature to mature neutrophil count ratio** > 0.2

Urine

Urinalysis and urine culture – indicated for late onset sepsis; not useful in early onset sepsis.

Töller score

Combined clinical and laboratory findings:

- < 5 – normal
- 5-10 – suspected
- > 10 points – definite sepsis

Other ancillary investigations

- **Neutrophil band to total neutrophil ratio** – normal < 0.16 (12 hr of life) and < 0.12 after 60 hr
- Elevated **micro erythrocyte sedimentation rate**
- Elevated **C-reactive protein**
- Elevated **high sensitivity CRP (hs-CRP)** $> 1 \text{ mg/L}$
- Elevated **procalcitonin** $> 2\text{-}2.5 \text{ ng/mL}$ (after postnatal 72 hours)
- Haptoglobin
- Buffy coat smear
- **Cytokines and chemokines** – CD11b, CD64, IL-6, IL-8 as markers of inflammation; also monokines, RANTES, MCP-1 and IP-10
- **Serum Amyloid A (SAA)**
- **Lipopolysaccharide Binding Protein (LBP)**

Other laboratory findings

- **Immature neutrophils** $> 10\%$
- **Immature:total neutrophil ratio** > 0.2
- **Thrombocytopenia** $< 100,000 \times 10^9/\text{L}$
- **CRP** $> 10 \text{ mg/L}$, or 2 SD above normal value
- **Procalcitonin** $> 8.1 \text{ mg/dL}$, or 2 SD above normal value
- **IL-6 or IL-8** $> 70 \text{ pg/mL}$
- **16S PCR positive**

Other investigations

- Polymerase chain reaction techniques to recover bacteria
- Latex agglutination tests
- Clotting profile

- Liver function test – to rule out bacterial hepatitis
- Blood glucose
- Electrolytes
- **Renal function – because of aminoglycosides**

Blood culture and other samples

- **Blood culture is positive in only 5-10% of neonatal sepsis**
- **Arterial blood gas** – indicated for signs or symptoms of hypoxia
- **Lumbar puncture** – indications include primary diagnosis of sepsis; positive blood culture; and neurological signs or symptoms

Superficial swab cultures

Cultures obtained from superficial regions – axilla, umbilical cord, outer ear canal, nasopharynx and orogastric tubes – show **poor correlation**.

Routine collection of superficial swab cultures is NOT recommended in neonatal sepsis.

Tracheal aspirate cultures are NOT recommended in prolonged intubation, due to rapid colonization following intubation.

Urine culture is NOT recommended in EOS.

Antibiotic therapy

Rational use of antibiotics: the correct use of quality antibiotics, in the right dosage, with wide microbial coverage, for the right duration, via the right mode of delivery, for the right patient.

Practices vary across Nigeria, ranging from **ampicillin/gentamicin combinations** through **2nd to 3rd generation cephalosporins**.

Nosocomial infections are often treated with **vancomycin or fluoroquinolones + amikacin**.

However, resistance of *Klebsiella* spp. to **gentamicin is about 60-72%**, and to **3rd generation cephalosporins 57-66%**.

Monitor response with: clinical state, CRP, procalcitonin, immature/total neutrophil levels.

Choice of initial antibiotic(s) is based on:

- **Sensitivity pattern in the local setting**
- Review of response of the patient, especially where laboratory evaluation is poor

WHO Committee Recommendation:

- **First line:** penicillin (benzylpenicillin/ampicillin/amoxicillin) + aminoglycoside (gentamicin), then vancomycin/cloxacillin + aminoglycoside (amikacin)
- **Second line:** penicillin/cephalosporin (cefotaxime/ceftriaxone) + aminoglycoside (amikacin)

Supportive management

- Correction of **anaemia, electrolyte and acid/base imbalances**
- **Prevent hypoxia and hypoglycaemia/hyperglycaemia**

- **Maintain a thermoneutral environment**
- **Exchange blood transfusion, buffy coat transfusion** – for sclerema, bacterial hepatitis with high conjugated bilirubin, leucopenia
- Treatment with **recombinant cytokines**
- **Management of hypotension and shock**

Other ancillary treatments

Treatments that increase the number or function of neutrophils – **granulocyte transfusions, GM-CSF, intravenous immune globulin (IVIG)** – are of **doubtful use** in neonatal sepsis.

Pentoxifylline, which works by **reducing TNF-alpha concentrations** associated with sepsis and **improving microcirculation**, showed **better survival**.

Prevention

- Minimise/reduce risk factors
- **Intrapartum penicillin prophylaxis 2-4 hours before SVD**, especially in culture-positive mothers for GBS – significantly reduces EOS
- **Good ANC** to avoid preterm/LBW and asphyxiated deliveries
- **Regular hand washing** to reduce skin/mucosa contamination
- **Eye care with 0.5% erythromycin eye ointment stat**
- **Cord care with 4% chlorhexidine gel once daily for 5-7 days**
- **Deliver baby on mother's abdomen for skin-to-skin contact for the critical 1 hour**
- **Early and exclusive breastfeeding**
- **Reduce traffic flow in the newborn unit**
- **Personalize facilities** among the newborns on admission
- Sepsis control in the newborn unit
- **Rational use of antibiotics**